

Journal of Applied Statistics

ISSN: 0266-4763 (Print) 1360-0532 (Online) Journal homepage: http://www.tandfonline.com/loi/cjas20

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To cite this article: Yuanhua Feng (2013) An iterative plug-in algorithm for decomposing seasonal time series using the Berlin Method, Journal of Applied Statistics, 40:2, 266-281, DOI: 10.1080/02664763.2012.740626

To link to this article: http://dx.doi.org/10.1080/02664763.2012.740626

Published online: 06 Nov 2012.

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Journal of Applied Statistics
Vol. 40, No. 2, February 2013, 266–281

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An iterative plug-in algorithm for decomposing seasonal time series using the Berlin Method

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(*Received 30 April 2012; final version received 13 October 2012*)

We propose a fast data-driven procedure for decomposing seasonal time series using the Berlin Method, the procedure used, e.g. by the German Federal Statistical Office in this context. The formula of the asymptotic optimal bandwidth h_A is obtained. Methods for estimating the unknowns in h_A are proposed. The algorithm is developed by adapting the well-known iterative plug-in idea to time series decomposition. Asymptotic behaviour of the proposal is investigated. Some computational aspects are discussed in detail. Data examples show that the proposal works very well in practice and that data-driven bandwidth selection offers new possibilities to improve the Berlin Method. Deep insights into the iterative plug-in rule are also provided.

Keywords: time series decomposition; Berlin Method; local regression; bandwidth selection; iterative plug-in

MSC2000 Codes: 62G08; 62G20; 62-07; 65Y15

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1. Introduction

Decomposing seasonal time series into unobserved components is an important issue of statistics. This question arises, if, e.g. we want to analyse monthly data or to build models using seasonally adjusted data. In this paper the equidistant additive time series model

$$Y_t = g(x_t) + S(x_t) + \epsilon_t, \quad t = 1, 2, \ldots, n, \tag{1}$$

is used to perform this, where $x_t = (t - 0.5)/n$, g is a smooth trend-cyclical component and S is a slowly changing seasonal component with period s. To simplify detailed discussion on bandwidth selection we assume in this paper that ϵ_t are iid random variables with $E(\epsilon_t) = 0$ and $\mathrm{var}(\epsilon_t) = \sigma^2$. The results can be easily extended to models with dependent errors. Model (1) can be treated as

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ISSN 0266-4763 print/ISSN 1360-0532 online
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http://dx.doi.org/10.1080/02664763.2012.740626
http://www.tandfonline.com

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a nonparametric regression with an additional (deterministic) seasonal component. Heiler [11] proposed to estimate g and S using local regression with polynomials and trigonometric functions as local regressors. This became the basis of the so-called Berlin Method (BV: Berliner Verfahren), which in its fourth version (BV4) is used, e.g. by the German Federal Statistical Office since 1983 [20,21]. A great advantage of BV4 is its mathematical clarity, which makes BV4 user-friendly [4]. Moreover, it allows the application of recent developments in modern nonparametric regression to further improve this approach.

A crucial problem of the use of BV is the selection of the bandwidth, as this procedure only performs better than related approaches if the bandwidth is suitably selected. Heiler and Feng [12] proposed to select the bandwidth under model (1) using a double-smoothing procedure. However, the running time of this procedure is very long, which hinders its application in practice. In this paper a very fast and practically relevant algorithm for selecting the bandwidth under model (1) is developed based on the iterative plug-in idea [8]. To our knowledge, this is the first detailed study on plug-in bandwidth selection for time series decomposition. Moreover, results of this paper also provide deep insights into the iterative plug-in idea and asymptotic behaviour of the proposal is investigated. Furthermore, some computational aspects are discussed in detail. Application to different data examples shows that the proposal works in principle very well in practice and that the data-driven bandwidth selection offers new possibilities to improve the Berlin Method.

The paper is organized as follows. The estimators and related properties are described in Section 2. Estimation of the unknowns for bandwidth selection is discussed in Section 3. The plug-in algorithm is proposed and studied in detail in Section 4. Data examples in Section 5 illustrate the practical usefulness of the proposal. Final remarks in Section 6 conclude the paper. Proofs of the results are put in the appendix.

2. The local regression approach

2.1 *The estimators*

Assume that g is at least $(p+1)$ times continuously differentiable, so that it can be expanded in a Taylor series around a point x_t. Similarly, S can be locally modelled by a Fourier series. Let $m = g + S$ denote the mean function. A general version of BV is defined as follows. Let $\lambda_1 = 2\pi/s$ be the seasonal frequency and $\lambda_j = j\lambda_1$, for $j = 2, \ldots, q$, where $q = [s/2]$ with $[\cdot]$ denoting the integer part. Let $K(u)$ be a second-order kernel function with compact support $[-1, 1]$. Let h denote the (half of the) bandwidth. The locally weighted regression estimators of g, S and m at x_t are obtained by solving the weighted least-square problem

$$Q = \sum_{i=1}^{n} \left\{ Y_t - \sum_{j=0}^{p} \beta_{1j}(x_i - x_t)^j - \sum_{j=1}^{q} (\beta_{2j} \cos \lambda_j (i-t) + [\beta_{3j} \sin \lambda_j (i-t)]) \right\}^2 K\left(\frac{x_i - x_t}{h}\right) \Rightarrow \min. \tag{2}$$

The solutions of Equation (2) are $\hat{g}(x_t) = \hat{\beta}_{10}$, $\hat{S}(x_t) = \sum_{j=1}^{q} \hat{\beta}_{2j}$ and $\hat{m}(x_t) = \hat{g}(x_t) + \hat{S}(x_t)$, where the coefficients and their estimators are defined locally and hence depend on x_t. For more details see [6,12].

Let

$$\mathbf{X}_1 = \begin{pmatrix} 1 & x_1 - x_t & \cdots & (x_1 - x_t)^p \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n - x_t & \cdots & (x_n - x_t)^p \end{pmatrix}$$

and

$$\mathbf{X}_2 = \begin{pmatrix} \cos\lambda_1(1-t) & \sin\lambda_1(1-t) & \cdots & \cos\lambda_q(1-t) & [\sin\lambda_q(1-t)] \\ \vdots & \vdots & \ddots & \vdots & [\vdots] \\ \cos\lambda_1(n-t) & \sin\lambda_1(n-t) & \cdots & \cos\lambda_q(n-t) & [\sin\lambda_q(n-t)] \end{pmatrix}.$$

Then $\mathbf{X} = (\mathbf{X}_1 \vdots \mathbf{X}_2)$ is the $n \times (p+s)$-design matrix. Entries in Equation (2) and $\mathbf{X}_2$ marked by [ ] only apply to odd s, for even s they have to be omitted due to $\lambda_q = \pi$. Let $\mathbf{y} = (y_1, \ldots, y_n)'$ be the observation vector and $\mathbf{K}$= diag(k_i) is a diagonal matrix with

$$k_i = K\left(\frac{x_i - x_t}{h}\right).$$

Furthermore, denote the jth $(p+1) \times 1$ unit vector by $\mathbf{e}_j$ and let Φ_s be an $(s-1) \times 1$ vector having 1 in its odd entries and 0 elsewhere. Then we have

$$\hat{m}(x_t) = (\mathbf{e}_1', \Phi_s')(\mathbf{X}'\mathbf{K}\mathbf{X})^{-1}\mathbf{X}'\mathbf{K}\mathbf{y} =: \mathbf{w}'\mathbf{y}, \tag{3}$$

$$\hat{g}(x_t) = (\mathbf{e}_1', \mathbf{0}')(\mathbf{X}'\mathbf{K}\mathbf{X})^{-1}\mathbf{X}'\mathbf{K}\mathbf{y} =: \mathbf{w}_1'\mathbf{y}, \tag{4}$$

and

$$\hat{S}(x_t) = (\mathbf{0}', \Phi_s')(\mathbf{X}'\mathbf{K}\mathbf{X})^{-1}\mathbf{X}'\mathbf{K}\mathbf{y} =: \mathbf{w}_2'\mathbf{y}, \tag{5}$$

where $\mathbf{0}$ is a vector of zeros of appropriate dimension.

The vectors $\mathbf{w} = (w_1, \ldots, w_n)'$, $\mathbf{w}_1 = (w_{11}, \ldots, w_{1n})'$ and $\mathbf{w}_2 = (w_{21}, \ldots, w_{2n})'$ are called weighting systems of $\hat{m}$, $\hat{g}$ and $\hat{S}$, respectively. We have $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$, $\sum w_i = \sum w_{1i} = 1$ and $\sum w_{2i} = 0$. The local regression approach makes $\hat{m}$, $\hat{g}$ and $\hat{S}$ exactly unbiased, if g is a polynomial of order not greater than p and S is exactly periodic with period s.

2.2 *Asymptotic properties*

In the sequel, it is assumed that p is odd so that $\hat{g}$ has automatic boundary correction. For the development of a plug-in bandwidth selector we need to discuss the asymptotic behaviour of $\hat{g}$, $\hat{S}$ and $\hat{m}$. Put $k = p + 1$ and assume that

(A1) $h \to 0$ and $nh \to \infty$ as $n \to \infty$.
(A2) g is at least k times continuously differentiable.
(A3) S is exactly periodic with period s.

(A1) and (A2) are the same as in nonparametric regression without seasonality. (A3) is only made to avoid the estimation of the bias in $\hat{S}$. However, model (1) works well in the case with slowly changing seasonality and a fixed selected bandwidth. Under (A1) it can be shown that $\hat{g}$ is asymptotically equivalent to some kernel estimator. Hence, the same asymptotic results in local polynomial fitting hold for $\hat{g}$ under model (1). The equivalent kernel for estimating g will be denoted by $K_p(u)$, which is of order k.

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To deal with $\hat{S}$, we will introduce a kernel estimator of S. Let

$$Q_s(i) = \begin{cases} (s-1) & \text{if } \dfrac{(i-t)}{s} \text{ is an integer,} \\ -1 & \text{otherwise,} \end{cases} \tag{6}$$

and

$$\check{w}_{2i} = (nh)^{-1} Q_s(i) K\left(\frac{x_i - x_t}{h}\right). \tag{7}$$

A kernel estimator of S is defined by

$$\check{S}(x_t) = \sum_{i=1}^{n} \check{w}_{2i} y_i =: \check{\mathbf{w}}_2' \mathbf{y}. \tag{8}$$

Note that $\{\check{w}_{2i}\}$ are asymptotically periodic with the same period s. Suppose that corresponding boundary correction is done for $\check{S}$, then it can be shown that $\hat{S}$ and $\check{S}$ are asymptotically equivalent under (A1) [6].

As an error criterion for bandwidth selection, the mean averaged squared error (MASE) is used. Define $R(K) = \int_{-1}^{1} K^2(u)\,\mathrm{d}u$. Let B denote the bias of an estimator. We have

LEMMA 1 *Assume that* (A1)–(A3) *hold, then*:

(1) *The asymptotic bias of* $\hat{m}$ *is*

$$B[\hat{m}(x_t)] \doteq B[\hat{g}(x_t)] \doteq \frac{1}{(k!)} \left\{ \left[\int u^k K_p(u)\,\mathrm{d}u \right] g^{(k)}(x_t) \right\} h^k. \tag{9}$$

(2) *The asymptotic variance of* $\hat{m}$ *is*

$$\mathrm{var}(\hat{m}(x_t)) = (nh)^{-1} \sigma^2 \{R(K_p) + (s-1)R(K)\}\{1 + O[(nh)^{-1}]\}. \tag{10}$$

(3) *The* MASE *of* $\hat{m}$ *is*

$$\begin{aligned} \mathrm{MASE}(\hat{m}) &:= \frac{1}{n} \sum_{t=1}^{n} [E(\hat{m}(x_t)) - m(x_t)]^2 \\ &\doteq \frac{\sigma^2}{nh} \{R(K_p) + (s-1)R(K)\} \\ &\quad + \frac{1}{(k!)^2} \left\{ \int \{g^{(k)}(x)\}^2\,\mathrm{d}x \left[\int u^k K(u)\,\mathrm{d}u \right]^2 \right\} h^{2k}. \end{aligned} \tag{11}$$

A sketched proof of Lemma 1 is given in the appendix, where it is shown in particular that: (1) $\hat{g}$ and $\hat{S}$ are asymptotically uncorrelated and (2) the bias in $\hat{S}$ is negligible compared with that in $\hat{g}$. The asymptotically optimal bandwidth, which minimizes the dominant part of the MASE is given by

$$h_{\mathrm{A}} = \left(\frac{(k!)^2}{2k} \frac{\sigma^2 \{R(K_p) + (s-1)R(K)\}}{\int \{g^{(k)}(x)\}^2\,\mathrm{d}x \{\int u^k K_p(u)\,\mathrm{d}u\}^2} \right)^{1/(2k+1)} n^{-1/(2k+1)}, \tag{12}$$

where it is assumed that $I = \int \{g^{(k)}(x)\}^2\,\mathrm{d}x > 0$. The change in h_{A} due to S is just an additional term $(s-1) * R(K)$ in the kernel-dependent constant of the variance of $\hat{m}$. For $s = 1$, the above formulae reduce to known results in nonparametric regression [5,18].

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3. Estimating the unknown parameters

3.1 *Estimation of the variance*

In order to develop a plug-in bandwidth selector based on Equation (12), the unknowns σ^2 and I have to be estimated. It is well known that the variance in nonparametric regression can be estimated by difference-based methods [9,10,16]. This idea can be extended to seasonal-difference-based variance estimators under model (1) [12]. Here, a sequence $D_{ms} = \{d_j \,|\, j = 0, 1, \ldots, m\}$ is called a *seasonal difference sequence*, if

$$\sum_{j=0}^{m} d_j = 0, \quad \sum_{j=0}^{m} d_j^2 = 1, \ \ m = 1, 2, \ldots \tag{13}$$

and

$$\sum_{j=0}^{m} d_j \delta_{ij} = 0, \quad i = 0, 1, \ldots, s-1, \tag{14}$$

where

$$\delta_{ij} = \begin{cases} 1 & \text{if } \dfrac{(j-i)}{s} \text{ is an integer,} \\ 0 & \text{otherwise.} \end{cases}$$

A seasonal-difference-based variance estimator is then defined by

$$\hat{\sigma}_{\mathrm{D}}^2 = (n-m)^{-1} \sum_{i=1}^{n-m} \left(\sum_{j=0}^{m} d_j Y_{i+j} \right)^2. \tag{15}$$

Following [10], it can be shown that $\hat{\sigma}_{\mathrm{D}}^2$ is root n consistent under (A2) and (A3). In this paper, the following seasonal difference sequence

$$D_{m,s} = \tfrac{1}{12}\{-1, 2, -1, \underbrace{0, \ldots, 0}_{s-3}, 1, -2, 1\}$$

defined for $s \geq 3$ will be used to estimate σ^2, where $m = s + 2$.

3.2 *Estimation of I*

Similar to local polynomial fitting, the kth derivative of g can be estimated with a local polynomial of order p_{I} and a bandwidth h_{I} with $p_{\mathrm{I}} > k$ and $p_{\mathrm{I}} - k$ odd. Here, $l = p_{\mathrm{I}} + 1$ is set. A simple choice is $p_{\mathrm{I}} = k + 1$ with $l = k + 2$. Now, let Equation (2) be defined with p being replaced by p_{I}. Let $\mathbf{K}$, $\mathbf{y}$ and $\mathbf{e}_j$ be the same as defined in Section 2. Let $\mathbf{X}$ be defined similarly as before. Then $\hat{g}^{(k)} = k!\hat{\beta}_k$ estimates $g^{(k)}$, which is given by

$$\hat{g}^{(k)}(t) = k!(\mathbf{e}_{k+1}', \mathbf{0}')(\mathbf{X}'\mathbf{K}\mathbf{X})^{-1}\mathbf{X}'\mathbf{K}\mathbf{y} =: (\mathbf{w}^k)'\mathbf{y}, \tag{16}$$

where $\mathbf{0}$ is the same as in Equation (4) and $\mathbf{w}^k = (w_1^k, \ldots, w_n^k)'$ is the weighting system of $\hat{g}^{(k)}$. I may then be estimated by

$$\hat{I}[g^{(k)}(x; h_{\mathrm{I}})] = n^{-1} \sum_{i=1}^{n} \{\hat{g}^{(k)}(x_i; h_{\mathrm{I}})\}^2. \tag{17}$$

In the following, some results on $\hat{I}$, which are important for the development of a plug-in bandwidth selector, will be given without proof, since we are only interested in the magnitude orders and these orders are the same for models with or without seasonality.

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Assume now that

(A1′) $h \to 0$ and $nh^{2k+1} \to \infty$ as $n \to \infty$.
(A2′) g is at least l times continuously differentiable.

Under Assumptions (A1)′, (A2)′ and (A3) we have

$$B(\hat{I}) \doteq O(h_{\mathrm{I}}^{(l-k)}) + O[(nh_{\mathrm{I}})^{-1}h_{\mathrm{I}}^{-(2k)}] \tag{18}$$

and

$$\mathrm{var}(\hat{I}) \doteq O(n^{-1}) + O(n^{-2}h_{\mathrm{I}}^{-4k-1}). \tag{19}$$

(A1)′ implies that h_{I} is of a larger order than h_{A}, i.e. $(h_{\mathrm{I}})^{-1} = o[(h_{\mathrm{A}})^{-1}]$, which ensures that $\hat{g}^{(k)}$ and hence $\hat{I}$ are at least consistent. See [17] for related results in nonparametric regression without seasonality. The following remarks show how h_{I} should be chosen.

Remark 1 The largest order that h_{I} should take is $O(n^{-1/(4k+2)}) = O[(h_{\mathrm{A}})^{1/2}]$. Under this choice the second term on the right-hand side of Equation (18) and the standard deviation of $\hat{I}$ achieve the fastest root n convergence rate at the same time. An h_{I} of an even larger order will increase the bias without improving the variance (in terms of the magnitude order).

Remark 2 The optimal bandwidth for estimating $g^{(k)}$ itself is of order $O(n^{-1/(2l+1)})$. This order is smaller than that in Remark 1, but larger than that in Remark 3. The choice $h_{\mathrm{I}} = O(n^{-1/(2l+1)})$ is hence also reasonable.

Remark 3 Observe that the MSE (mean squared error) of $\hat{I}$ is dominated by the squared bias part. By balancing the orders of the two terms on the right-hand side of Equation (18) we obtain $h_{\mathrm{I}} = O(n^{-1/(k+l+1)})$, which may be considered to be the (asymptotically) optimal choice of h_{I}. This order is smaller than both orders mentioned above.

4. The main proposal

4.1 *The basic algorithm*

Hereafter, only $p = 1$ and 3 with $k = 2$ and 4 will be considered. Following the iterative plug-in idea of Gasser *et al.* [8], $\hat{I}_j$, the estimate of I in the jth iteration, is calculated with a bandwidth $h_{\mathrm{I},j}$, which is obtained from h_{j-1}, the bandwidth for estimating m in the $(j-1)$th iteration, by means of an inflation method. Here, an inflation method is a function $h_{\mathrm{I},j} = f(h_{j-1})$ such that $(h_{\mathrm{I},j})^{-1} = o[(h_{j-1})^{-1}]$. Hence, $h_{\mathrm{I},j}$ will be of a larger order than h_{A}, if h_{j-1} is at least of order $O(h_{\mathrm{A}})$. Now (A1)′ is satisfied so that $\hat{I}$ and $\hat{h}$ will be both consistent in the jth iteration. Two inflation methods will be considered.

The original idea of Gasser *et al.* [8], called a multiplied inflation method (MIM), is to set $h_{\mathrm{I},j} = f(h_{j-1}) = ch_{j-1}n^{\alpha}$ with some $\alpha > 0$, called the inflation factor. This idea is discussed in detail by Herrmann and Gasser [13]. There are some unknowns in the function f such as c, α and a starting bandwidth h_0, which have to be fixed beforehand. The rate of convergence of $\hat{h}$ does not depend on c and h_0. In this paper, we will simply choose $c = 1$. The choice of h_0 will be discussed in Section 4.3. Let $l = k + 2$. Following Remarks 1–3, we have three reasonable choices of α for the MIM, respectively:

(1) $\alpha_1 = 1/(4k+2)$ so that the variance term of $\hat{I}$ is minimized,
(2) $\alpha_2 = 4/[(2k+1)(2k+5)]$ so that $\hat{g}^{(k)}$ is optimized and

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(3) $\alpha_3 = 2/[(2k+1)(2k+3)]$ so that the MSE of $\hat{I}$ is minimized,

when convergence is reached, where $\alpha_1 > \alpha_2 > \alpha_3$ and α_3 is the asymptotically optimal choice of α.

It is well known that the required number of iterations (J^0, say) by the MIM is very large, especially for $k > 2$. For example, if $k = 4$, it is $J^0 = 5k + 1 = 21$ for α_1 and $J^0 = (k+1)(2k+1) = 45$ for α_3 [13]. Beran and Feng [1] introduced another inflation method $h_{\mathrm{I},j} = f(h_{j-1}) = ch_{j-1}^{\beta}$, called an exponential inflation method (EIM). This idea is studied by Beran and Feng [2] in detail. They show that in order to inflate h_{A} to a given order, the required number of iterations by the EIM is much smaller than by the MIM. In the following, the EIM with $c = 1$ will hence be used. Following [2], the choices of β corresponding to α_1, α_2 and α_3 above are:

(1) $\beta_1 = \frac{1}{2}$,
(2) $\beta_2 = (2k+1)/(2k+5)$ and
(3) $\beta_3 = (2k+1)/(2k+3)$,

where $\beta_1 < \beta_2 < \beta_3$ and β_3 is the asymptotically optimal choice of β.

In the following, we will propose a basic iterative plug-in algorithm for selecting the bandwidth in time series decomposition, which is defined for $k = 2$ and $k = 4$ separately.

(i) Start with a possible bandwidth h_0.
(ii) For $j = 1, 2, \ldots$ set $h_{\mathrm{I},j} = h_{j-1}^{\beta}$ with $\beta = \beta_3 = \frac{5}{7}$ for $k = 2$ and $\beta = \beta_2 = \frac{9}{13}$ for $k = 4$. Calculate

$$h_j = \left(\frac{(k!)^2}{2k} \frac{\hat{\sigma}^2 \{R(K_p) + (s-1)R(K)\}}{\int \{\hat{g}^{(k)}(x; h_{\mathrm{I},j})\}^2 \, \mathrm{d}x \{\int u^k K_p(u) \, \mathrm{d}u\}^2} \right)^{1/(2k+1)} n^{-1/(2k+1)}. \tag{20}$$

(iii) Increase j by 1 and repeat Step (ii) until convergence is reached at some j^0 and set $\hat{h} = h_{j^0}$.

For related plug-in bandwidth selectors in nonparametric regression without seasonality see [8,13,14,17].

Theoretically, β_3 is the asymptotically optimal choice of β. Our experience shows that for $k = 2$, this choice works well in practice. Hence, we choose $\beta_3 = \frac{5}{7}$ for $k = 2$. However, $\beta_3 = \frac{9}{11}$ for $k = 4$ is too close to one and for small samples the bandwidth could not be inflated correctly. For $k = 4$, it is hence proposed to use the slightly stronger inflation factor β_2. Now, the variance of $\hat{h}$ with $k = 2$ and $k = 4$ is almost of the same order and $\hat{h}$ is hence in both cases stable (see Theorem 1 in the next subsection). The most stable inflation factor $\beta_1 = \frac{1}{2}$ by the EIM is too strong and does not work well for small samples.

4.2 *Asymptotic behaviour*

The iterative plug-in algorithm is motivated by fixed point search. Here, the procedure is started with a bandwidth h_0 and stopped, if a convergent output (a fixed point) is achieved. The inflation process behind an iterative plug-in algorithm is described by the following lemma according to the relationship between h_0 and h_{A}.

LEMMA 2 *Under assumptions* (A2)′ *and* (A3), *an iterative plug-in algorithm processes as follows:*

Case 1. *Start with an* $h_0 = o_p(h_{\mathrm{A}})$, *then*

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Step 1. $h_j = O_p(h_{\mathrm{I},j})$, *if* $h_{\mathrm{I},j} = o_p(h_{\mathrm{A}})$.
Step 2. $h_j = O_p(h_{\mathrm{A}})$, *if* $h_{\mathrm{I},j} = O_p(h_{\mathrm{A}})$.
Step 3. $h_j = h_{\mathrm{A}}[1 + o_p(1)]$, *if* $h_{\mathrm{A}} = o_p(h_{\mathrm{I},j})$.
Case 2. *Start with an* h_0 *such that* $(h_0)^{-1} = o_p[(h_{\mathrm{A}})^{-1}]$, *then*
Step 1′. $h_j = O_p(h_{\mathrm{A}})$, *if* $h_{\mathrm{I},j} = O_p(1)$.
Step 2′. *The same as Step 3 in case 1.*

The proof of Lemma 2 is given in the appendix. Related results may be found in Herrmann and Gasser [13, p. 8] and Beran and Feng [2]. Note in particular that (A1′) is not required by Lemma 2.

Starting with a small bandwidth, Case 1 in Lemma 2 shows that h_{j-1} will be inflated in the jth iteration, if $h_{j-1} = o_p(h_{\mathrm{A}})$. This will be repeatedly carried out until $h_{j'} = O_p(h_{\mathrm{A}})$ is reached in the j'th iteration and $h_{j'+1}$ in the next iteration will be a consistent bandwidth selector. Some further iterations are required to improve the finite sample property of $\hat{h}$.

Case 2 in Lemma 2 shows how the estimation procedure of such an algorithm works, if a starting bandwidth h_0, which is at least of order $O_p(h_{\mathrm{A}})$, is used. On the one hand, if $h_0 = o_p(1)$, then h_1 is already consistent, since (A1′) is satisfied. In this case, Step 1′ will not appear. On the other hand, if $h_0 = O_p(1)$, then $h_1 = O_p(h_{\mathrm{A}})$, which is already of the correct order but not yet consistent. However, h_2 will be consistent. Again, some further iterations are required to reduce the influence of h_0.

The following theorem holds for the algorithm proposed in Section 4.1.

THEOREM 1 *Under the assumptions of Lemma* 2 *we have*

(i) *For* $k = 2$ *with* $\beta_3 = \frac{5}{7}$

$$\hat{h} = h_{\mathrm{A}}\{1 + O(n^{-2/7}) + O_p(n^{-5/14})\}. \tag{21}$$

(ii) *For* $k = 4$ *with* $\beta_2 = \frac{9}{13}$

$$\hat{h} = h_{\mathrm{A}}\{1 + O(n^{-2/13}) + O_p(n^{-9/26})\}. \tag{22}$$

A sketched proof of Theorem 1 is given in the appendix.

Let h_{M} denote the optimal bandwidth, which minimizes the MASE. Theorem 1 also holds, if h_{A} on the right-hand sides of Equations (21) and (22) is replaced by h_{M}. This is due to the fact that $|h_{\mathrm{M}} - h_{\mathrm{A}}|/h_{\mathrm{M}} = O(h_{\mathrm{M}}^2)$ [3], which is of orders $O(n^{-2/5})$ for $k = 2$ and $O(n^{-2/9})$ for $k = 4$, and is hence negligible. Furthermore, the advantage of a plug-in bandwidth selector compared with a double-smoothing bandwidth selector is that it runs very fast. But the rate of convergence of a plug-in bandwidth selector is usually slower than a corresponding double-smoothing bandwidth selector [7]. This disadvantage is not that grave, because a slight change in the rate of convergence of a bandwidth selector will not affect the goodness-of-fit of the resulting nonparametric regression estimators as much.

4.3 *Computational aspects*

This section deals with computational aspects such as the choice of j^0, the choice of h_0 and so on. A more practical procedure will be proposed at the end of this section.

The estimators in Section 2.1 are defined with a fixed bandwidth h. In this case, the number of observations used at x_t decreases when x_t moves from the interior to the boundary. To solve this problem the k-NN idea will be used. For a given h, we define a left bandwidth h_{l} and a right one h_{r} so that $h_{\mathrm{l}} = h_{\mathrm{r}} = h$ in the interior, $h_{\mathrm{l}} = x_t$ at a left boundary point and $h_{\mathrm{r}} = 1 - x_t$ at a right

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boundary point. h_r (or h_l, respectively) at a boundary point is then determined by $h_l + h_r = 2h$. The estimates at a boundary point are calculated similarly but with h in Equation (2) being replaced by $\max(h_l, h_r)$.

In our software only bandwidths $h \in [h_{\min}, h_{\max}]$ with $h_{\min} = s/n$ and $h_{\max} = 0.5 - 1/n$ will be considered, which includes practically all reasonable possibilities of h. Furthermore, two bandwidths h and h' will be considered to be the same, if $|h - h'| < 1/n$, because a difference of such an order is for any bandwidth selector negligible. In the software, the bandwidth actually used is an integer $b_h = [nh + 0.5]$, which is the (half of the) bandwidth w.r.t. the observation time t. The total number of observations used at each time point is $N_h = 2b_h + 1$. Let $b_{h_{1,j}} = [nh_{1,j} + 0.5]$. Then we obtain a natural criterion for stopping the computing procedure, i.e. the procedure will be stopped, if $b_{h_{1,j^0}} = b_{h_{1,j^0-1}}$ in the j^0th iteration. This implies that $\hat{I}_j^0 = \hat{I}_{j^0-1}$ and $\hat{h} = h_{j^0} = h_{j^0-1}$. Further iterations are not necessary. Note that even the j^0th iteration is just a repetition of the $(j^0 - 1)$th iteration.

In the following, the choice of h_0 will be considered. In most cases, h_0 does not play any role. However, in some cases, when the finite sample MASE has more than one local minima or when the MASE changes very slowly around its unique minimum, then $\hat{h}$ may depend on h_0 in some way. To explain this we will introduce some concepts. A bandwidth h_f is called a fixed point (of the procedure proposed in Section 4.1), if $\hat{h} = h_f$, when the procedure is started with $h_0 = h_f$ itself. A fixed point h_f is called left stable, if for all $h_0 \le h_f$ in a neighbourhood of h_f we have $\hat{h} = h_f$. A fixed point h_f is called right stable, if for all $h_0 \ge h_f$ in a neighbourhood of h_f we have $\hat{h} = h_f$. A fixed point h_f is called stable, if it is both left and right stable. A fixed point is called unstable, if it is only achievable by starting with itself. An interval of bandwidths $[h_f^l, h_f^r]$ is called an interval of fixed points, if h_f^l is a left stable fixed point, h_f^r is a right stable fixed point and all points between them are unstable fixed points. Denote by $\hat{h}^l$ the bandwidth selected with $h_0^1 = h_{\min}$ and by $\hat{h}^r$ the bandwidth selected with $h_0^2 = h_{\max}$. Then $\hat{h}^l$ is a left stable fixed point, if $\hat{h}^l > h_{\min}$ and $\hat{h}^r$ is a right stable fixed point, if $\hat{h}^r < h_{\max}$.

When the finite sample MASE has only one minimum, then a unique stable fixed point or a unique interval of fixed points exits. In the first case, we will obtain the same selected bandwidth $\hat{h}$ by starting with any h_0. In the second case, we have $\hat{h} = h^l$ for all $h_0 \le h^l$, $\hat{h} = h^r$ for all $h_0 \ge h^r$ and $\hat{h} = h_0$ for $h^l < h_0 < h^r$. Now, all bandwidths in $[h^l, h^r]$ are reasonable to be used as the optimal bandwidth, since the change of the MASE over $[h_f^l, h_f^r]$ is negligible. In this case we also say that the result is *unique*, which will be set at $\hat{h} := (\hat{h}^l + \hat{h}^r)/2$. In the following, *a stable fixed point* can also refer to an interval of fixed points. In the case when the finite MASE has more than one local minima, the selected bandwidth by starting with different h_0 may be different. Now, there may also be some unstable fixed points corresponding to a local maximum between two local minima. If this is the case, we should find out all possible stable fixed points and then select one of them as the bandwidth to use by analysing the smoothing results further.

An S-Plus® function called DeSeaTS (Decomposing Seasonal Time Series) is developed based on the following quasi-data-driven procedure.

(1) Carry out the algorithm in 4.1 twice with $h_0^1 = h_{\min}$ and $h_0^2 = h_{\max}$, respectively.
(2) Calculate the decomposition results automatically, if $\hat{h}$ is unique.
(3) Show detailed information about all stable fixed points, when $\hat{h}$ is not unique.

If (3) occurs, further subjective analysis is required.

For choosing p, we propose to carry out the above procedure with $p = 1$ and $p = 3$, respectively. If the smoothing results with $p = 1$ and $p = 3$ are both satisfactory, we can choose either $p = 1$ or $p = 3$. However, it is more preferable to use $p = 3$, since now the selected bandwidth is in general slightly larger, which does not increase the bias of $\hat{g}$ but will improve $\hat{S}$. Sometimes one choice of

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p is more reasonable than the other, thus the reasonable one should be chosen (see the examples given in the next section). An objective criterion for choosing p is not given here, because we do not have an estimate of the MASE at the end of the procedure.

5. Practical performance

The following data examples are chosen to show the practical performance of the proposal.

(1) The Series 'CAPE' – time series of the quarterly final consumption expenditure in Australia (total private, millions of dollars, 1989/90 prices) from September 1959 to June 1995 with $n = 144$. Source: Australian Bureau of Statistics.
(2) The Series 'Strom' – the monthly time series of produced electricity in Germany from 1955 to 1979 with $n = 300$. Source: [19, p. 82].
(3) The Series 'IFOR' – the monthly time series of the indices of the foreign orders received in Germany from 1978 to 1994 (1985 = 100) with $n = 204$. Source: IFO-Institute for Economic Research in Munich.
(4) The Series 'Hsales' – monthly sales of new one-family houses sold in the USA from January 1973 to November 1995 with $n = 275$. Source: Makridakis *et al.* [15].

All of these time series are analysed with $p = 1$ and $p = 3$, respectively. Throughout the application, the bisquare kernel is used. The selected bandwidths and the number of iterations with the smallest starting bandwidth $h_0^1 = h_{\min} = s/n$ and the largest starting bandwidth $h_0^2 = h_{\max} = 0.5 - 1/n$, together with the answer, whether the two bandwidths are the same, are listed in Table 1 for all data examples. From Table 1 we see that the two selected bandwidths in most of the cases are unique. For the series Hsales with $p = 3$, we obtained an interval of fixed points [0.094, 0.105]. As mentioned before, we will consider such a result to be *unique* and now $\hat{h} = (0.105 + 0.094)/2 = 0.10$ will be used. Two unusual cases should be mentioned: First, the selected bandwidths for the series IFOR with $p = 1$ are not unique. Second, although the selected bandwidth for the series Strom with $p = 3$ is unique, it is, however, much smaller than that selected for the same series with $p = 1$. This means that the proposal does not work well for the Strom series with $p = 3$. For the final smoothing we hence propose to use $p = 1$ for the Strom series and $p = 3$ for the others. Data-driven decomposition results for these examples are shown in Figure 1(a)–(d), where corresponding location changes are introduced for the seasonal component so that the figures look more clear. We see that the results given in Figure 1 look quite well. This shows the practical usefulness of the proposed procedure. Note that the selected bandwidths for the examples given in Figure 1(a)–(d) are quite different, which adapt automatically to the structure of the data. The largest bandwidth is $\hat{h} = 0.16$ by the series Strom. This is not surprising because the trend in this time series can almost be modelled by a parametric model [19]. Although the trend in the time series CAPE is also regular, the selected bandwidth $\hat{h} = 0.089$ is the smallest

Table 1. $\hat{h}^{\mathrm{l}}$, $\hat{h}^{\mathrm{r}}$ and other parameters for the data examples.

	$p = 1$					$p = 3$				
Time series	$\hat{h}^{\mathrm{l}}$	j^0	$\hat{h}^{\mathrm{r}}$	j^0	Uniq.	$\hat{h}^{\mathrm{l}}$	j^0	$\hat{h}^{\mathrm{r}}$	j^0	Uniq.
CAPE	0.084	7	0.086	6	Yes	0.089	6	0.089	8	Yes
Strom	0.160	7	0.160	7	Yes	0.101	7	0.102	13	Yes
IFOR	0.113	6	0.262	3	No	0.140	7	0.141	6	Yes
Hsales	0.066	4	0.067	8	Yes	0.094	7	0.105	4	Int

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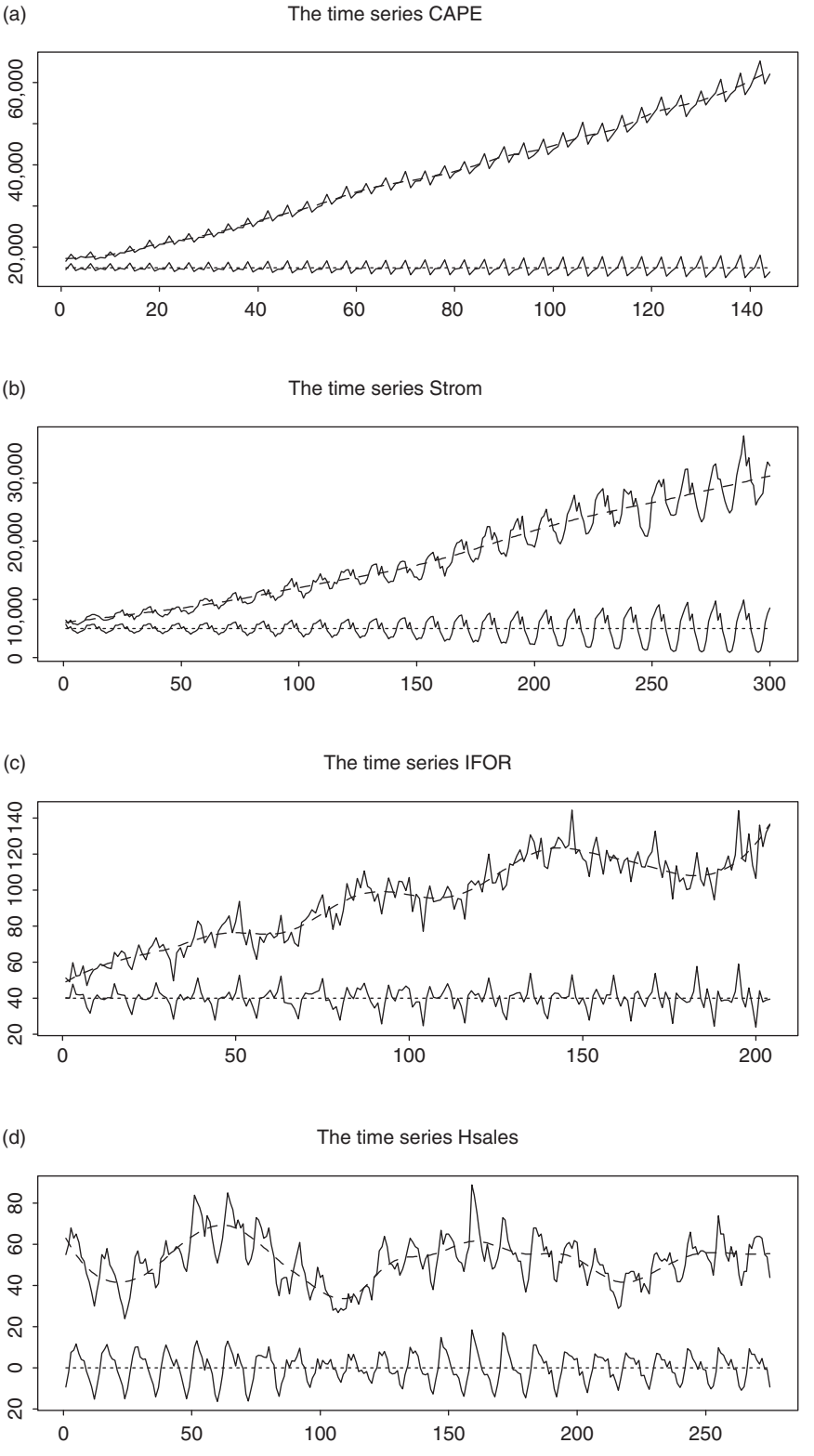


Figure 1. Optimal decomposition results for the data examples. Upper: the data together with the estimated trend (dashes). Below: the estimated seasonal component.

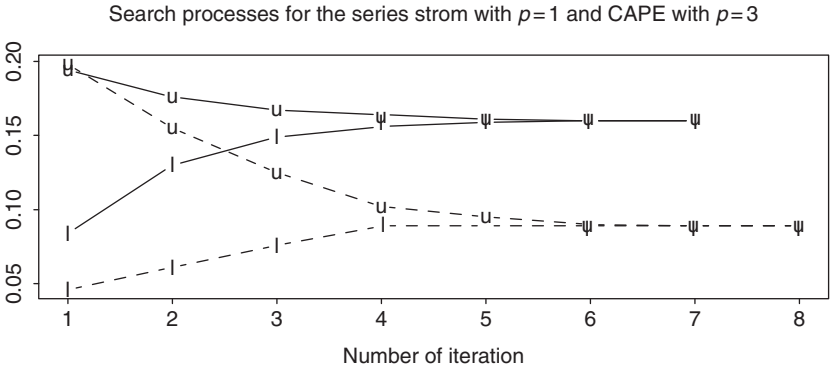


Figure 2. Search processes for Strom ($p = 1$, solid line) and CAPE ($p = 3$, dashed line). The letters 'l' and 'u' indicate results with $h_0^1 = h_{\min}$ and $h_0^2 = h_{\max}$, respectively.

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one, since $s = 4$ for this time series but for the other $s = 12$. Table 1 also shows that j^0 changes from case to case.

Furthermore, it is easy to calculate that the above bandwidths for the four examples correspond to 27 quarters, and 97, 57 and 59 months, respectively. The selected bandwidths in the first two cases are much larger than the bandwidths used in the current version of BV4 (BV4.1) [20,21]. Possibly, the performance of BV4.1 can be improved, if it is combined with the proposed bandwidth selection algorithm.

Finally, we want to show some detailed properties of the iterative plug-in algorithm so that the reader can understand the proposal well. Following Lemma 2, we have $\hat{h}^{\mathrm{l}} \leq h_{\mathrm{A}} \leq \hat{h}^{\mathrm{r}}$ in probability. From Table 1 we see that this is true for all examples. Lemma 2 also ensures that, in probability, h_j is nondecreasing in j by starting with h_0^1 and h_j is nonincreasing in j by starting with h_0^2. The detailed search processes with starting bandwidths h_0^1 and h_0^2, respectively, are shown in Figure 2, where the results are for the time series Strom with $p = 1$ (solid line) and CAPE with $p = 3$ (dashed line). From Figure 2 we can see that the selected bandwidth for Strom with $p = 1$ is much larger than that for CAPE with $p = 3$; h_1 with h_0^2 in the second case is however slightly larger than that in the first case. Still, after a few iterations both of them achieve their corresponding fixed points.

6. Final remarks

This paper proposes an iterative plug-in algorithm for decomposing seasonal time series using the Berlin Method. Computational aspects of the proposal are discussed in detail. A few useful properties of the iterative-plug-in rule were found. Data examples show that the proposal works very well in practice. The facts that the selected bandwidths vary from one series to another very strongly and that all of the selected bandwidths are clearly larger than the default bandwidth used in BV4.1 should stimulate studies testing whether the integration of a suitable bandwidth selector will improve the quality of the procedure. This study is the first detailed study on bandwidth selection for decomposing economic time series. Possible extensions are left open for research involve, for instance, the proper combination of the proposed algorithm with BV4.1, the adaptation of the algorithm according to the dependence structure of the errors, bandwidth selection under consideration of the bias in $\hat{S}$ and selection of two separate bandwidths for estimating g and S, respectively.

Acknowledgements

This work was partly supported by the *Center for International Economics*, University of Paderborn. We are very grateful to the Editor and two referees for their useful comments and suggestions, which helped to improve the quality of the paper clearly. The data for the time series CAPE and Hsales are downloaded from the *Time Series Data Library*. We would like to thank Prof. Rob J. Hyndman, Monash University, for making these data publicly available.

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Appendix. Proofs of the results

A sketched proof of Lemma 1: The proof of this lemma based on some desirable standardizing and orthogonal finite sample properties of $\hat{g}$ and $\hat{S}$. These properties are quantified by the following

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properties of $\mathbf{w}_1$ and $\mathbf{w}_2$.

$$\begin{aligned}
&\text{a.} \quad \sum_{i=1}^{n} w_{1i}(x_i - x_t)^j = \begin{cases} 1, & j = 0, \\ 0, & 1 \le j \le p, \end{cases} \\
&\text{a}'. \quad \begin{cases} \sum_{i=1}^{n} w_{1i} \cos(\lambda_j(i-t)) = 0, \\ \sum_{i=1}^{n} w_{1i} \sin(\lambda_j(i-t)) = 0, \end{cases} \qquad j = 1, \ldots, q. \\
&\text{b.} \quad \sum_{i=1}^{n} w_{2i}(x_i - x_t)^j = 0, \quad 0 \le j \le p, \\
&\text{b}'. \quad \begin{cases} \sum_{i=1}^{n} w_{2i} \cos(\lambda_j(i-t)) = 1, \\ \sum_{i=1}^{n} w_i \sin(\lambda_j(i-t)) = 0, \end{cases} \qquad j = 1, \ldots, q.
\end{aligned} \tag{A1}$$

Note that $\mathbf{w}_1' = (\mathbf{e}_1', \mathbf{0}')(\mathbf{X}'\mathbf{K}\mathbf{X})^{-1}\mathbf{X}'\mathbf{K}$ and $\mathbf{w}_2' = (\mathbf{0}', \Phi_s')(\mathbf{X}'\mathbf{K}\mathbf{X})^{-1}\mathbf{X}'\mathbf{K}$. Hence we have $\mathbf{w}_1'\mathbf{X} = (\mathbf{e}_1', \mathbf{0}')$ and $\mathbf{w}_2'\mathbf{X} = (\mathbf{0}', \Phi_s')$. Observing the definition of $\mathbf{e}_1$ and Φ_s we obtain the results in Equation (A1). Note that Equation (A1) ensures that $\hat{g}$, $\hat{S}$ and hence $\hat{m}$ are exactly unbiased, if m is the sum of a polynomial of order no larger than p and S is an exactly periodic component with period s.

(1) Under (A2) and (A3) we have, in the neighbourhood of x_t,

$$g(x) = \sum_{j=0}^{p} \frac{g^{(j)}(x_t)}{j!}(x - x_t)^j + \frac{g^{(k)}(x_t + \theta(x - x_t))}{k!}(x - x_t)^k, \tag{A2}$$

where $0 < \theta < 1$ and

$$S(x_i) = \sum_{j=1}^{q} (\beta_{2j} \cos \lambda_j(i-t) + [\beta_{3j} \sin \lambda_j(i-t)]). \tag{A3}$$

This leads to $S(x_t) = \sum_{j=1}^{q} \beta_{2j}$. Following a′, we have

$$B[\hat{g}(x_t)] = \sum_{i=1}^{n} w_{1i}[g(x_i) + S(x_i)] - g(x_t) = \sum_{i=1}^{n} w_{1i} g(x_i) - g(x_t), \tag{A4}$$

since $\sum_{i=1}^{n} w_{1i} S(x_i) = 0$. For $B(\hat{S})$ we have

$$B[\hat{S}(x_t)] = \sum_{i=1}^{n} w_{2i}[g(x_i) + S(x_i)] - S(x_t) = \sum_{i=1}^{n} w_{2i} g(x_i), \tag{A5}$$

since $\sum_{i=1}^{n} w_{2i} S(x_i) = \sum_{j=1}^{q} \beta_{2j} = S(x_t)$ following b′ and Equation (A3). Property a′ results in

$$\sum_{i=1}^{n} w_{2i} \left\{ \sum_{j=0}^{p} \frac{g^{(j)}(x_t)}{j!}(x - x_t)^j \right\} = 0. \tag{A6}$$

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Hence

$$B[\hat{S}(x_t)] = \sum_{i=1}^{n} w_{2i} \frac{g^{(k)}(x_t + \theta(x_i - x_t))}{k!}(x_i - x_t)^k$$
$$\doteq \frac{g^{(k)}(x_t)}{k!} h^k \sum_{i=1}^{n} w_{2i} \left(\frac{x_i - x_t}{h}\right)^k$$
$$= o(h^k), \tag{A7}$$

where the last equation is due to the fact

$$\sum_{i=1}^{n} w_{2i} \left(\frac{x_i - x_t}{h}\right)^{k'} = o(1), \text{for any} k' \geq 0. \tag{A8}$$

Equation (A8) holds, since the weights w_{2i} are asymptotically periodic (see Equation (7)). This shows that $B(\hat{S})$ is only due to the kth order term in the Taylor expansion of g. Furthermore, the contribution of this term to $B(\hat{S})$ is negligible compared with $B(\hat{g})$. We obtain

$$B[\hat{S}(x_t)] = o(B[\hat{g}(x_t)])$$

and

$$B[\hat{m}(x_t)] \doteq B[\hat{g}(x_t)].$$

Observe that $B(\hat{g})$ is the same as for a local polynomial fitting of order p, we obtain Equation (9).

(2) Detailed proof of Equation (10) may be found in Feng [6], where is it shown in particular that the two weighting systems $\mathbf{w}_1$ and $\mathbf{w}_2$ are asymptotically orthogonal in the sense that $\sum_{i=1}^{n} w_{1i} w_{2i} = o(\sum_{i=1}^{n} w_{1i}^2) = o(\sum_{i=1}^{n} w_{2i}^2)$. This follows from Equation (A8), since $K_p(u)$ is a polynomial kernel.

(3) Formula (11) follows from (9) and (10). Lemma 1 is proved.

In the following, it will be explained, why Lemma 2 and Theorem 1 should hold. Detailed proofs are omitted, since these results are similar to those in nonparametric regression without seasonality.

A sketched proof of Lemma 2: Case 1. Note that the two terms on the right-hand side of Equation (18) are due to the contribution of $B(\hat{g}^{(k)})$ and $\text{var}(\hat{g}^{(k)})$ (see, e.g. the proof of Proposition 1 in Beran and Feng [1]). In step 1 we have $h_{\text{I},j} = o(h_{\text{A}})$ in the jth iteration. In this case $B(\hat{g}^{(k)})$ is negligible and $\hat{I}$ is dominated by $\text{var}(\hat{g}^{(k)})$, which tends to infinite as $n \to \infty$. Observe that $w_i^k = O[(nh_{\text{I},j}^{k+1})^{-1}]$, we have $\text{var}(\hat{g}^{(k)}) = O(n^{-1} h_{\text{I}j}^{-(2k+1)})$ and hence $\hat{I} = O_p(n^{-1} h_{\text{I}j}^{-(2k+1)})$. Inserting this in the formula for h_j we obtain $h_j = O_p(h_{\text{I}j})$, i.e. in this case h_{j-1} is inflated to a bandwidth of order $O_p(h_{\text{I}j})$. Step 1 is proved. Results in Steps 2 and 3 are clear.

Case 2. Note that Step 1′ will not appear, if h_0 is of a larger order than h_{A} such that $h_0 \to 0$, since now (A1′) is satisfied in the first iteration. In this case h_1 is already consistent and only Step 2′ will appear. Step 1′ occurs, only if $0 < h_0 < 0.5$ is taken to be a constant. Now, $B(\hat{I}_1)$ is a constant and $\hat{I}_1 = O_p(1) = O_p(I)$. We hence obtain $h_1 = O_p(h_{\text{A}})$, which is of the correct order but not yet consistent. The process will then be changed into Step 2′ in the second iteration. Lemma 2 is proved.

Remark A1 Theoretically, if the procedure is started with an h_0 such that $h_{\text{A}} = o(h_0)$ and $h_0 \to 0$ as $n \to \infty$, then h_1 will already be consistent. Hence such a starting bandwidth is asymptotically more preferable. Now the asymptotic behaviour of an iterative plug-in bandwidth selector is easy to understand. If the sample size is small and the data have a special structure, a too large

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starting bandwidth, e.g. $h_0^2 = h_{\max}$ may perhaps lead to $\hat{I}_1 \doteq 0$. Now h_j could not be deflated to the optimal bandwidth. In the application we did not yet find such a phenomenon. If this occurs, it is no problem for our proposal, because it will be discovered by starting with the other bandwidth h_0^1.

A sketched proof of Theorem 1: The proof of Theorem 1 can be carried out based on a formula given in the appendix in [1], see also [2]. They showed that, when convergence is reached, the rate of convergence of an iterative plug-in bandwidth selector is quantified by:

$$\frac{(\hat{h} - h_{\mathrm{A}})}{h_{\mathrm{A}}} \doteq -\frac{1}{2k + 1 - 2\delta} I^{-1}(\hat{I} - I). \tag{A9}$$

Equation (A9) shows that $B(\hat{h})$ and $\mathrm{var}(\hat{h})$ at the end of the proposed procedure are of the corresponding orders as those of $\hat{I}$. $\mathrm{var}(\hat{h})$ is dominated by the second term in Equation (19) of order $O(n^{-1}h_{\mathrm{I}}^{-4k-1})$, where h_{I} denotes the bandwidth for estimating I used at the end of the procedure, which is of order $O_p(n^{-1/7})$ for $k = 2$ and $O_p(n^{-1/13})$ for $k = 4$. In both cases, i.e. $k = 2$ with β_3 and $k = 4$ with β_2, the order of the second term on the right-hand side of Equation (18) is no larger than that of the first. Hence we have $B(\hat{h}) = O[B(\hat{I})] = O(h_{\mathrm{I}}^{-2})$. Straightforward calculation leads to the results of Theorem 1.

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